

On-device artificial intelligence agent based on language models for electrochemical water desalination

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HIGHLIGHTS

- Edge-deployed LLM integrates electrochemical knowledge for control and prediction.
- Validated against 320 published desalination studies.
- Hybrid LLM-machine learning enables robust prediction under missing sensor inputs.
- Vision-language retrieval reduces hallucinations with full reference accuracy.
- Enables privacy-preserving, low-cost decentralized automation.

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ABSTRACT

Electrochemical water treatment is essential for tackling global water scarcity but remains difficult to optimize due to limited expertise and computing resources at many treatment facilities. Here, we introduce an intelligent on-device platform that combines electrochemical process knowledge with large language models deployed directly on edge devices such as Raspberry Pi. This system integrates theoretical understanding with real-time optimization, eliminating the need for cloud connectivity while ensuring data privacy and accessibility. Tested against 320 published studies, it achieves a 60 % reduction in hallucination rates and maintains high predictive accuracy ($R^2 > 0.80$) for key variables such as effluent concentration and energy, even with incomplete sensor inputs. Notably, prediction accuracy for challenging parameters, such as the applied current (the driving force for electrochemical water desalination), improves from 0.03 to 0.63. By bringing intelligence to the data rather than sending it to the cloud, this approach makes advanced water-treatment intelligence feasible in resource-limited, data-imperfect, decentralized environments where physics-based models cannot be deployed.

1. Introduction

Global freshwater gaps are widening at an accelerating pace and could grow by >15 % under high warming scenarios, placing billions at risk as climate change intensifies droughts and degrades existing water infrastructure. Meeting this challenge demands scalable, energy-efficient treatment technologies that can operate across diverse, resource-constrained settings. Electrochemical water treatment (ECWT) has emerged as a compelling alternative to conventional thermal and pressure-driven desalination, offering high ion selectivity, lower specific energy consumption, and direct compatibility with renewable electricity sources. Its modular architecture makes ECWT particularly well-suited

to decentralized applications where grid reliability and technical expertise are limited (Alkhadra et al., 2022; Barazesh et al., 2015; Guyes et al., 2021; Xu et al., 2024).

Despite these advantages, translating ECWT potential into reliable operational practice remains difficult. Key experimental parameters, including electrode architecture, operating protocols, feedwater chemistry, and performance metrics, are inconsistently reported across studies, complicating reproducibility and fair cross-study comparison (Łukaszczyk and Burda, 2024; Xu et al., 2025a). This challenge is compounded by an inherently high-dimensional, tightly coupled design space, where material properties, device configuration, control strategies, and water matrix composition interact nonlinearly. As highlighted

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in recent perspectives on digital water management (Xu et al., 2025b), there is a pressing need for a unified platform capable of organizing ECWT knowledge, linking experimental findings to their original evidence, and supporting transparent, evidence-based operational decision-making.

Conventional machine learning approaches have shown promise for narrowly defined ECWT optimization tasks when curated, complete datasets are available (Kim et al., 2024; Lee et al., 2025; Ullah et al., 2025). However, they frequently fail under missing-data conditions, struggle to generalize beyond their training distributions, and rarely provide traceable explanations for their predictions, limitations that substantially reduce their practical reliability in field deployments where sensor noise and data gaps are routine (Chen et al., 2025; Xu et al., 2025b). A more robust approach is to build systems that draw on accumulated domain knowledge to support prediction and control even under imperfect input conditions. More broadly, conventional physics-based and AI-based models provide physically grounded predictions under complete, clean input conditions (Ullah et al., 2025; Yoon et al., 2022a). However, three operational scenarios remain unaddressed by such models: (i) incomplete or missing sensor data, which occurs routinely due to sensor failure, fouling, or maintenance; (ii) deployment on resource-constrained edge devices without the computational infrastructure needed to run iterative solvers; and (iii) autonomous integration of distributed knowledge from the scientific literature for real-time decision support. These are precisely the conditions under which an AI-driven framework can offer complementary and practical value.

Recent advances in large language models (LLMs) and retrieval-augmented generation (RAG) have demonstrated remarkable potential for structured knowledge extraction and scientific reasoning across domains, from medicine to biodiversity research (Agarwal et al., 2024a,b; Bednarczyk et al., 2025; Kommineni et al., 2024a, 2024b; Nejjar et al., 2023; Poornima et al., 2024; Sarker, 2021; Scherbakov et al., 2025; Thirunavukarasu et al., 2023). RAG connects a language model to a searchable index of domain documents; when queried, it first retrieves the most relevant passages from text, tables, and figure captions, and then composes an answer grounded in those citations. Vision-language systems extend this further by processing full-document PDFs, capturing meaning encoded in figures, tables, and layouts, and extracting structured experimental variables with near-expert precision. When combined with fine-tuned LLMs, this capability enables translation of heterogeneous literature into actionable process insights (Sarker, 2021; Shi and Ding, 2025; Yoon et al., 2022a, 2022b). Concurrently, model quantization techniques have reduced LLM memory footprints by approximately fourfold while improving inference speed, making deployment on low-power edge hardware increasingly feasible and opening new possibilities for real-time, off-grid process optimization (Chen et al., 2025).

Building on these developments, we introduce an on-device, vision-enabled expert-agent platform that autonomously gathers, analyzes, and applies electrochemical desalination knowledge to support process understanding and operational control. Using membrane capacitive deionization (MCDI) as a representative ECWT technology, the system integrates three interconnected layers of intelligence: (i) a vision-language retrieval framework that answers theoretical and data-specific queries by extracting and contextualizing evidence from a large ECWT literature corpus; (ii) a hybrid predictive control module that translates grounded knowledge into quantitative process predictions and operating recommendations, even under missing input conditions; and (iii) a portable, resource-efficient deployment architecture enabling fully offline, real-time decision support at the point of treatment. Together, these components form a unified framework that links scientific literature, numerical process modeling, and operational control, demonstrating that localized, explainable AI for water treatment optimization is not only feasible but also practical for next-generation decentralized water systems.

2. Methods

2.1. Literature corpus and data pre-processing

A corpus of 320 peer-reviewed ECWT papers (2006–2025) with emphasis on MCDI was assembled (Fig. S1). The literature search was conducted using Google Scholar and Web of Science with the query: “electrochemical desalination optimization; process design; modelling; control.” Non-research and non-English documents were excluded, and duplicates were removed through manual screening. Metadata, including paper titles and author names, was extracted using Python scripts to enable accurate referencing for responses generated by the RAG system.

2.2. Structured data generation for MCDI processes

To fully assess the effectiveness of the AI agent for electrochemical water desalination systems on both structured and unstructured data, a dataset consisting of 6000 data points was generated using an experimentally validated modified Donnan model (Text S1–S2). This model, referenced in prior published studies (Ullah et al., 2025; Yoon et al., 2022a), was selected due to its robust capability to simulate dynamic fouling effects by employing an empirical correlation between operation time and fouling thickness for accurate estimation of fouling accumulation on the electrode surface (Text S1, Eq. S1). The underlying working principle and stepwise operational workflow of the advanced modified Donnan model are systematically depicted through a block diagram (Text S2, Fig. S7). The advanced modified Donnan model employed here accounts for multi-ion transport, explicitly incorporating Na^+ , Ca^{2+} , and Cl^- ion fluxes through the membrane and electrode (Text S2, Eqs. S3–S17). The inlet concentration range (18.24–19.75 mM) represents brackish water conditions relevant to decentralized treatment applications. Critically, the model incorporates dynamic fouling effects through an empirical fouling thickness correlation calibrated against experimental data (Text S1, Eq. S1), electrode area reduction due to fouling (Eqs. S18–S19), and microporosity degradation over time (Eqs. S20–S21), enabling the 6000 training data points to capture realistic long-term performance degradation, in contrast to idealized models that assume clean electrode conditions.

Data were generated using a combination of a random sampling function and the differential evolution optimization algorithm, enabling datasets that represent both optimized and unoptimized process conditions for the MCDI system. The numerical model inputs included current, flow rate, flushing intensity, voltage, and influent concentration, all varied within defined operational ranges (Table S1). The operational ranges used for structured data generation reflect typical MCDI operating conditions reported in the peer-reviewed literature, where similar orders of magnitude of current, flow rate, voltage, and feed salinity have been examined in laboratory and pilot-scale desalination studies (e.g., multichannel MCDI performance studies exploring feed flow rate and voltage as key parameters (Kim et al., 2021); pilot-scale desalination under constant-current MCDI operation examining continuous operation and feed salinities (Zhu et al., 2024); flexible tuning of current and flow to manage desalination productivity (He et al., 2024)). These examples support that the 6000 simulated data points represent physically realistic scenarios for MCDI systems. For the optimized dataset, the target effluent concentration was fixed at 17.0 mM (target water quality), aligning with the established drinking water quality standard (Ullah et al., 2025). The random dataset, however, did not impose a target concentration; the input variables were randomly varied across their respective ranges, and the corresponding output metrics from the numerical model were recorded.

2.3. Expert-level agents based on retrieval-augmented search (vision-language)

Documents are visually rich structures that convey meaning not only through text but also through figures, page layouts, tables, and even fonts. Traditional retrieval systems, which rely on extracting only text from pages through lengthy and fragile processes like OCR or layout parsing, often fail to capture these visual cues, limiting their usefulness in applications such as RAG. To address this challenge, the ColPali v1.0 (Vidore/colpali-v1.0) model was adopted as the retrieval backbone because it directly embeds entire document pages in their visual form, producing fine-grained representations that can align naturally with user queries (Faysse et al., 2024). The Byaldi Python library (Version 0.07) was used to streamline indexing and querying so that pages could be searched without additional pre-processing overhead. Each query

entered the same representation space and was compared to document pages using a ColBERT-style late interaction mechanism, allowing precise alignment between query terms and the corresponding text regions, figures, or table elements on a page (Fig. 1a). Through preliminary tests, ColPali was chosen because it faithfully captures both textual and visual information, distinguishes between documents that may look similar at a glance, and operates efficiently by skipping the time-consuming pipeline steps of conventional methods. Together, these qualities make ColPali especially well-suited for searching across diverse materials ranging from narrative scientific articles to structured data tables.

2.4. Design and implementation of the hybrid expert-level agent (EA_{hybrid})

A fine-tuning process was performed on the Mistral-7B-v0.1 language model to adapt it for handling structured data (Fig. 1b).

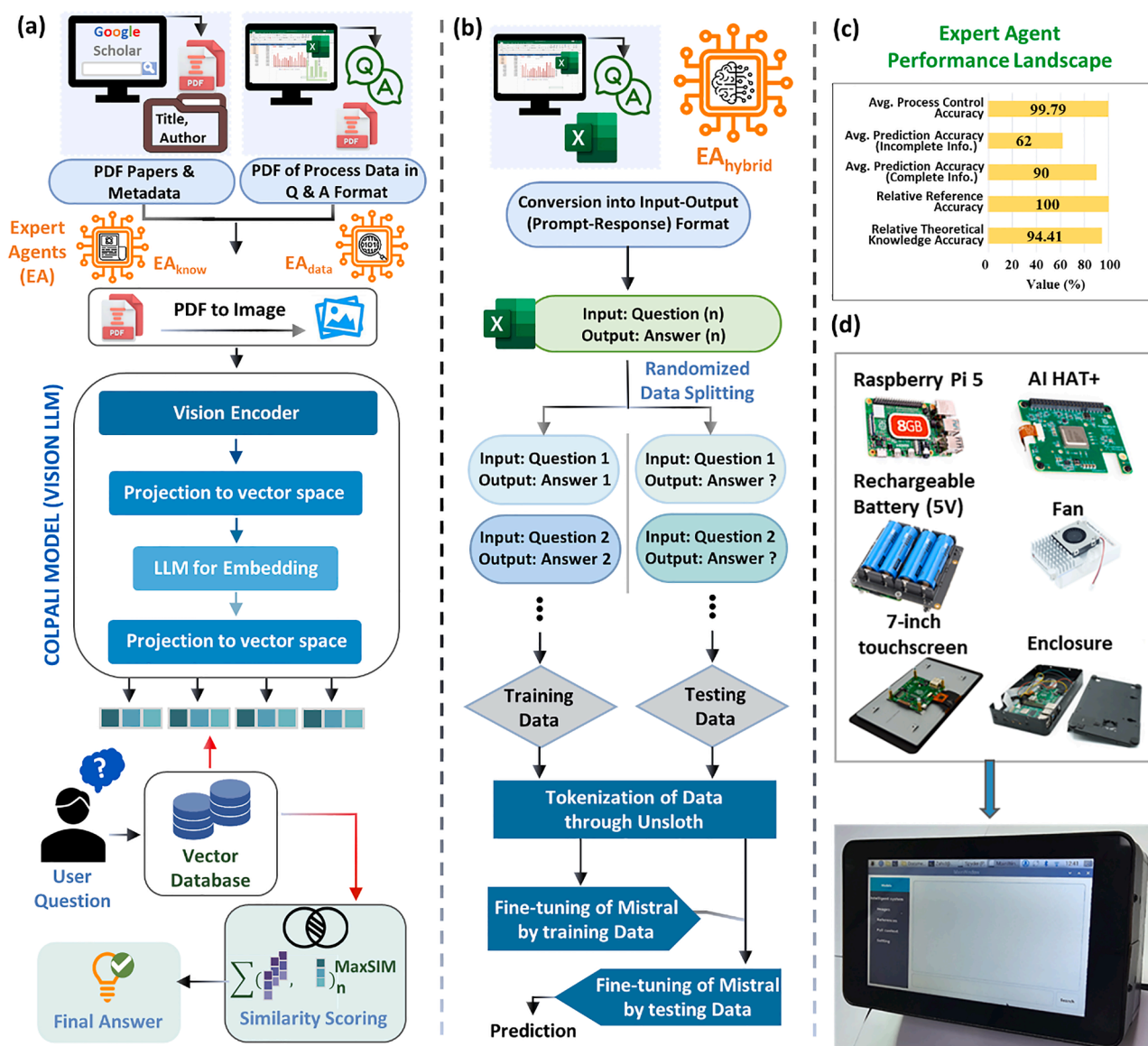


Fig. 1. Proposed framework to develop an on-device artificial intelligence-based expert agent for electrochemical desalination processes. (a) The knowledge retrieval component (EA_{know} and EA_{data}) using vision-language models. The ColPali vision LLM processes PDF documents (scientific papers and structured process data) to create vector embeddings, which are stored in a vector database. User queries are matched against this database using similarity scoring to retrieve relevant information and generate responses. (b) Predictive component (EA_{hybrid}) using fine-tuned language models. Structured process data are converted into input-output prompt-response format, then split into training and testing datasets for fine-tuning the Mistral language model to enable prediction, process control, and optimization. (c) Performance metrics of the expert agent across theoretical knowledge, reference accuracy, prediction accuracy (with complete and incomplete information), and process control accuracy. (d) Proof-of-concept deployment on a lightweight, Raspberry Pi-based portable system comprising a Raspberry Pi 5, AI HAT+, battery (5 V), fan, 7-inch touchscreen, and enclosure. The system operates fully offline without internet connectivity.

Approximately 6000 simulated structured data samples were used for this purpose (Table S2). The fine-tuning was conducted using the Unsloth library (October-2025 release), which provides an efficient way to work with large language models. The model, pre-quantized in 4-bit precision to reduce memory use, was loaded along with its default tokenizer, with a maximum sequence length set to 2048 tokens and automatic data type detection. Fine-tuning was carried out using the SFT Trainer class, which managed the training over the dataset with a batch size of 2 per device and gradient accumulation steps to simulate a total batch size of 8. Training included 500 steps with a fixed learning rate (2×10^{-4}), optimized by AdamW using 8-bit quantization to further reduce resource consumption. Mixed precision training was enabled depending on hardware compatibility (either fp16 or bf16). During training, 0.6 % of the model's parameters (about 42 million out of 7 billion) were updated, focusing on efficient adaptation. Finally, after fine-tuning, the model was quantized again and saved in the GGUF format for a smaller storage footprint and easier deployment. Based on preliminary tests of various language models, Mistral-7B was chosen for its superior performance and efficiency, outperforming larger models in reasoning, mathematics, and code tasks, while its advanced attention mechanisms enable faster inference and handling of long sequences, making it well

suited for real-world structured data applications (Jiang et al., 2023). This approach allowed fast and resource-effective customization of Mistral-7B for structured data tasks, leveraging the Unsloth framework's speed and low-overhead training capabilities.

2.5. Development of the on-device AI agent for cross-platform deployment

The software interface was created utilizing the Qt Designer tool (PyQt5 version 5.15.11) to establish a user-friendly graphical layout, followed by the implementation of functionality in Python (Fig. 1d). To facilitate on-device and standalone operation, the Python code was compiled into an executable application using the auto-py-to-exe library. This application integrates three expert system layers: a vision-based intelligent knowledge system for delivering reference-supported technical guidance; a structured-data analytics layer for analyzing plant data stored in Excel files and other tabular formats; and a hybrid intelligent layer for predictive modeling, decision support, and advanced process control in operational environments. The software was deployed on a Raspberry Pi 5 with 8 GB RAM and AI HAT+, which offers an embedded computing platform with adequate processing power and connectivity options. The Raspberry Pi was connected to a

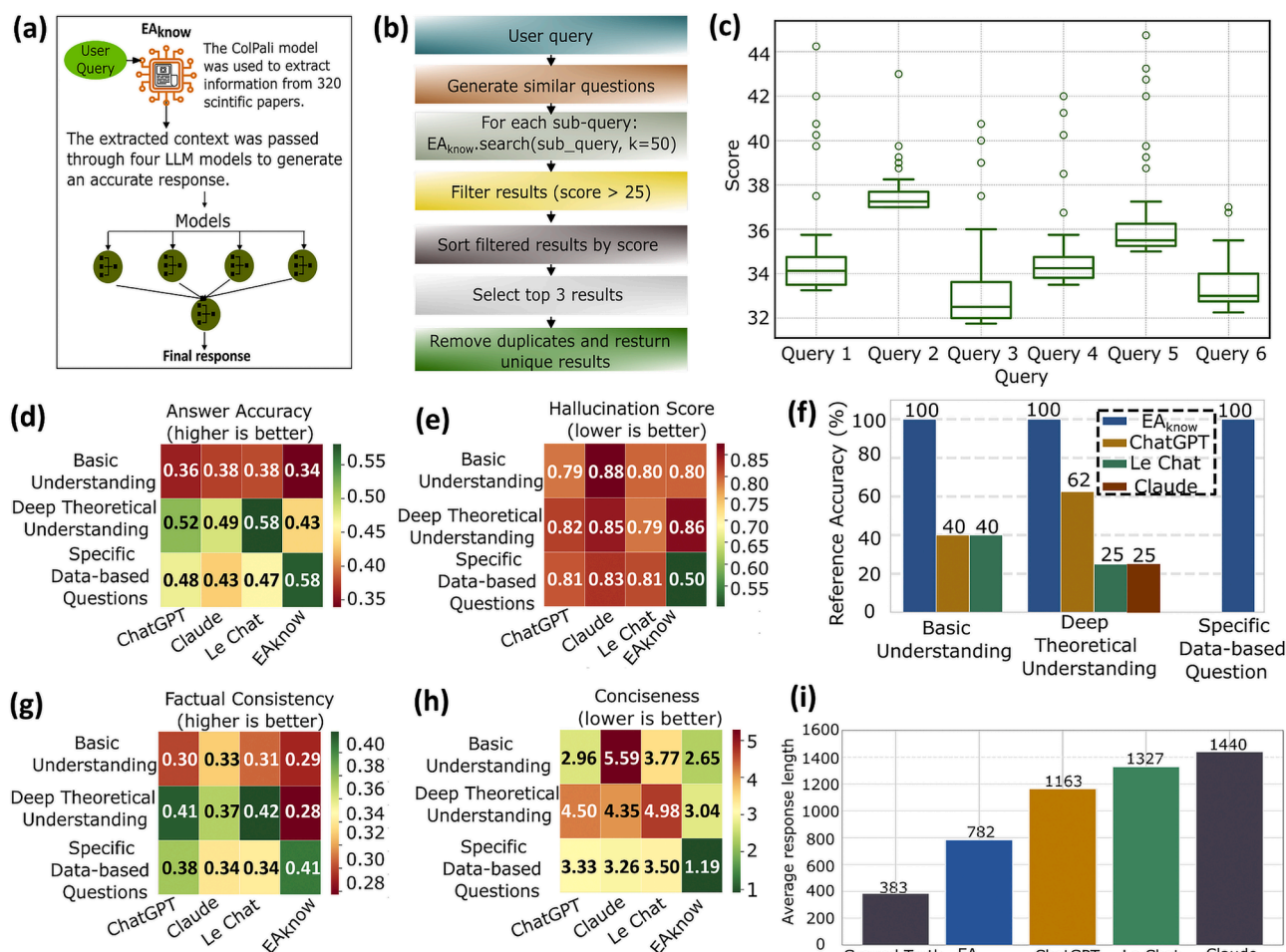


Fig. 2. Performance of the expert-level agent for scientific literature analysis (EAknow) of MCDI processes. (a) System workflow for answering user queries, combining ColPal model retrievals with an ensemble of LLM models. Four LLMs process the retrieved context independently, and their outputs are consolidated into a final response. (b) Retrieval pipeline showing query expansion, filtering, and ranking of candidate documents. Each user query generates similar sub-queries, searches the corpus (top 50 results per sub-query), filters by score threshold (>25), sorts, and selects the top three unique results. (c) Distribution of retrieval scores for six representative queries, showing consistent high median scores (34–38) indicating stable query-document alignment. (d–i) Comparative evaluation metrics: (d) answer accuracy, (e) hallucination score, (f) reference accuracy, (g) factual consistency, (h) conciseness, and (i) average response length. EAknow is benchmarked against GPT-5.1 (ChatGPT), Claude Sonnet 4, and Le Chat (November 2024 version) across three question categories: basic understanding, deep theoretical understanding, and specific data-based questions. Ground truth values are shown for reference. The Python evaluation pipeline and dataset for the EAknow benchmarking framework are available in the Zenodo repository.

compact 7-inch touchscreen display to enable direct, intuitive interaction. This integration of hardware and software provides a portable, self-sufficient solution that functions effectively without the need for an external internet connection.

2.6. Performance metrics for benchmarking and evaluation

To evaluate the theoretical knowledge performance of all question-answering models, including vision-based RAG systems, GPT-5.1 (accessed via ChatGPT; hereafter ChatGPT), Claude Sonnet 4 (hereafter Claude), and Le Chat by Mistral (November 2024 version; hereafter Le Chat), all models were tested using the same standardized MCDI question dataset with verified ground-truth answers derived from peer-reviewed literature. For each question, identical prompts were provided to all systems, and the generated responses were automatically processed using Python scripts. Text normalization, tokenization, and stop-word removal were performed using the NLTK library, and each response was compared with the corresponding ground-truth answer using quantitative evaluation metrics. The performance scores were computed for each question and then averaged across question types to produce the benchmarking results presented in Fig. 2. The evaluation framework included Exact Match, Answer Accuracy, Hallucination Score, Reference Accuracy, Factual Consistency, and Conciseness to assess different aspects of response quality. For numerical prediction tasks, Root Mean Square Error (RMSE) and coefficient of determination (R^2) were used to evaluate fine-tuned Mistral regression performance. All metrics were computed automatically without manual intervention, ensuring reproducibility and objective comparison, and each metric captures a different dimension of alignment between generated responses and verified scientific ground truth.

Answer Accuracy (AA): Quantifies the similarity between prediction and ground truth by first converting the texts into numerical vectors using Term Frequency-Inverse Document Frequency (TF-IDF), which reflects how important a word is to a document relative to a collection, followed by computing the cosine similarity between these vectors to measure their closeness. In simple terms, this metric compares the important words in the ground-truth answer and predicted answer and measures how close they are in meaning. The text is converted into numerical word vectors and their similarity is calculated; higher values indicate stronger agreement with the correct scientific answer. (Eq. (1)):

$$AA = \frac{\vec{y} \cdot \vec{y}}{\|\vec{y}\| \|\vec{y}\|} \quad (1)$$

where $\vec{y}$, $\vec{y}$ represent the TF-IDF vector representations of the ground truth and prediction, respectively.

Hallucination Score (HS): Estimates the proportion of unique tokens in the prediction that do not appear in the ground truth, after removing stop words and non-alphanumeric tokens. This metric counts how many extra or unsupported words appear in the predicted answer that do not exist in the ground truth. A lower hallucination score means the model introduces fewer incorrect or fabricated statements. (Eq. (2)):

$$HS = \frac{|T_{\hat{y}} - T_y|}{|T_{\hat{y}}|} \quad (2)$$

where $T_y, T_{\hat{y}}$ are the sets of meaningful tokens in the ground truth and prediction, respectively.

Reference Accuracy (RA): Shows the percentage of references that were both valid and relevant. Each reference is manually verified by checking whether it exists in scientific literature and whether the generated answer is supported by that reference. It is calculated by dividing the number of correct references by the total number of references and multiplying by 100 % (Eq. (3)):

$$RA = \frac{N_{correct}}{N_{total}} \times 100 \quad (3)$$

where $N_{correct}$ is the number of correct references and N_{total} is the total number of references.

Factual Consistency (FC): Measures the fraction of key phrases (n-grams of length 1 to 3) found in the prediction relative to those in the ground truth. Higher values reflect greater factual overlap (Eq. (4)):

$$FC = \frac{|N(y) \cap N(\hat{y})|}{|N(y)|} \quad (4)$$

where $N(y)$ represents the set of n-grams in the ground truth and $N(\hat{y})$ represents the set of n-grams in the prediction.

Conciseness Score (CS): Assesses the length ratio of the prediction compared to the ground truth, where 1 indicates equal length, values below 1 indicate more concise predictions, and values above 1 indicate verbosity (Eq. (5)):

$$CS = \frac{Length(\hat{y})}{Length(y)} \quad (5)$$

where $Length(\hat{y})$ is the number of tokens in the prediction and $Length(y)$ is the number of tokens in the ground truth.

Average Response Length (AL): The average response length was calculated for both the ground truth and each system under evaluation (Eq. (6)):

$$AL_s = \frac{1}{N} \sum_{i=1}^N length(y_i) \quad (6)$$

where s represents the target system—ground truth, N is the number of responses, and y is the response of the system.

Root Mean Square Error (RMSE): The RMSE provides a measure of the average magnitude of prediction error. Lower values indicate better predictive performance (Eq. (7)).

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (7)$$

where y_i is the observed (actual) value, $\hat{y}_i$ is the predicted value, and N is the number of observations.

Coefficient of Determination (R^2): The R^2 score measures how well the predicted values approximate the ground truth values. A score of 1 indicates perfect prediction (Eq. (8)):

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (8)$$

where $\bar{y}$ is the mean of the observed (actual) values.

3. Results

3.1. The expert-level agent for analyzing scientific knowledge (EA_{know})

To create an expert-level agent for ECWT scientific knowledge (EA_{know}), we adopted RAG for domain knowledge learning. We demonstrate the system using MCDI as a representative ECWT technology. The vision-based EA_{know} system demonstrates consistent improvements in accuracy, reliability, and efficiency for scientific literature analysis (Fig. 2). Its architecture (Fig. 2a) integrates page embeddings (ColPali-based model) with an ensemble of LLM models from Ollama, including Mistral, Math-Mistral, Openchat, and two domain-specific models from ALIENTELLIGENCE (Chemicalengineer and Environmentalengineer). The ColPali model extracts embeddings from a large corpus of scientific papers, routes them to the four LLMs, and a final model refines and consolidates their output into a single

coherent response.

The retrieval workflow (Fig. 2b) illustrates how ColPali extracts relevant information based on user queries. Each query expands into semantically related sub-queries, for which ColPali retrieves the top 50 document candidates. These are then filtered by threshold score, ranked, and the top three retained after removing duplicates. Fig. 2c displays retrieval score distributions, where high median values (34–38) indicate stable alignment between user input and retrieved documents. In ColPali, these scores are calculated by the model's built-in similarity function, which compares query embeddings with page embeddings (Fig. 1a).

For MCDI EA_{know} development, 320 scientific papers on MCDI processes were used as the core knowledge base (Fig. S1a). MCDI was selected for its balance of energy efficiency, ion selectivity, and low fouling, outperforming conventional capacitive deionization and providing advantages in sustainability and ion removal compared to electrodialysis and other ECWT methods (He et al., 2024). To evaluate the system's reasoning capabilities, three types of questions were generated: basic understanding (40 %), deep theoretical understanding (32 %), and specific data-based inquiries (28 %), as summarized in Fig. S1b. All evaluation questions were manually designed from peer-reviewed MCDI publications by systematically extracting factual statements, numerical values, mechanisms, and theoretical explanations from different sections of the literature (abstract, methodology, results, and discussion). Each question was paired with a verified ground-truth answer and corresponding literature reference to ensure traceability and scientific accuracy. The evaluation dataset was designed to cover different levels of scientific reasoning, including basic understanding, deep theoretical analysis, and data-based numerical interpretation. To ensure transparency and reproducibility, the complete Python evaluation pipeline and dataset for the EA_{know} benchmarking framework are available in the Zenodo repository. Evaluations highlighted significant advantages of the MCDI EA_{know} system. For answer accuracy (Eq. (1)), EA_{know} achieved 0.43 for deep theoretical understanding, comparable to ChatGPT (0.52), Claude (0.49), and Le Chat (0.58) (Fig. 2d). However, it showed a clear advantage for data-based questions, which require precise extraction of numerical values, parameter ranges, or tabulated experimental results (e.g., electrode capacitance, salt removal, or operating voltages). For this category, EA_{know} achieved accuracy of 0.58, higher than ChatGPT (0.47), Claude (0.43), and Le Chat (0.48). These gains in data-specific querying were accompanied by a major reduction in hallucination rates (Eq. (2)), with EA_{know} at 0.50, while all others exceeded 0.80, underscoring stronger grounding in retrieved documents (Fig. 2e). This reliability extended to reference attribution (Eq. (3)), where EA_{know} consistently achieved 100 % accuracy across all categories, unlike competing models, where reference accuracy fell as low as 0 % for data-based and basic understanding queries, and to 25 % for deep theoretical queries in the case of Le Chat and Claude (Fig. 2f). Factual consistency (Eq. (4)) also mirrored these trends (Fig. 2g). For specific data-driven questions, EA_{know} achieved 0.41, outperforming ChatGPT (0.38), Claude (0.34), and Le Chat (0.34). Beyond correctness, responses were notably concise (Fig. 2h). Conciseness scores (Eq. (5)) ranged from 1.19 to 3.04, aligning with the compact, context-rich style of expert answers, while other LLMs were consistently more verbose, scoring between 3.2 and 5.5. Average response length (Eq. (6)) (Fig. 2i) further reflected this, with EA_{know} producing responses averaging 782 tokens, significantly shorter than Claude (1440), Le Chat (1327), and ChatGPT (1163), and much closer to the ground-truth benchmark of 383 tokens. Overall, the vision-language EA_{know} system consistently outperformed other models across all metrics, particularly in retrieval and data-specific queries, making it a reliable and scalable framework for literature-driven scientific discovery in electrochemical desalination.

3.2. The expert-level agent for analyzing structured data (EA_{data}) of MCDI processes

Building upon the vision-language EA_{know} system developed for extracting and reasoning over the MCDI literature, the framework was extended to structured numeric data (Fig. 3). This extension addresses an industrial need where large spreadsheet datasets must be analyzed with precision, scalability, and speed.

A dataset of 6000 structured data points generated from the advanced modified Donnan model for MCDI was used; the model had been experimentally validated and cited in prior studies (Ullah et al., 2025; Yoon et al., 2022a). The dataset was created within specified operating ranges (Table S1), and a detailed summary is provided in Table S2. Since the ColPali model processes visually encoded documents, the structured data were reformatted into a question-answer schema (Table S3) and compiled into an extended PDF document for ingestion. The constructed queries targeted MCDI performance descriptors such as effluent concentration (mM), optimum operating conditions (mA or V), thermodynamic efficiency (%), and energy consumption (kWh m^{-3}).

Accurate adjustment of effluent water quality in response to inlet concentration changes is critical to MCDI automation. To assess this, the EA_{data} system analyzed six queries aimed at determining optimum operating conditions for specific effluent concentrations under given inlet concentrations (Fig. 3a). User queries were processed through the ColPali retrieval model, which searched a long PDF file to identify the closest or exact numerical match. Retrieved results were then passed to the LLM for structured response generation. Performance was evaluated by comparing actual values and extracted values for inlet concentration (C_{in}), effluent concentration (C_{eff}), current (I), flow rate (Q), flushing intensity (f), and cell voltage (V_{c}) (Fig. 3b–g). For example, Query 1 (Fig. 3b), with extracted values that reproduced the actual data well for concentrations, current, and flushing intensity, showed minor discrepancies in flow rate and cutoff voltage. Query 2 (Fig. 3c) showed strong consistency across all parameters, with small deviation in current, while Query 3 (Fig. 3d) showed excellent agreement except for flushing intensity, which exhibited a larger deviation. Subsequent evaluations (Fig. 3e–g) followed similar trends, consistently reproducing concentration parameters (C_{in} and C_{eff}) with high fidelity but showing small deviations in secondary operating variables (flow rate, flushing intensity, and voltage). EA_{data} retrieved critical parameters effectively, though minor errors appeared where information was less explicit, confirming its robustness for both scientific papers and structured datasets, and its suitability for industrial process optimization. We also performed the same analysis for queries containing a longer formatted dataset (four decimal points). While the extraction trends persisted, we observed slightly larger deviations for less explicit variables, supporting the selection of two-decimal-point queries for practical reporting (Fig. S2). However, the system depends strongly on prompt formulation—it cannot predict unseen information. Different prompt designs could shift focus and outputs, highlighting prompt engineering as a key step in adapting EA_{data} systems to specific domains and applications.

3.3. The hybrid model that combines fine-tuned language model and machine learning (EA_{hybrid}) is suitable for process automation

To develop the expert system for prediction, control, and optimization that is capable of making predictions even with incomplete input, a large language model (Mistral-7B model) was fine-tuned on 6000 structured data points, generated from the MCDI numerical model (Table S2). This fine-tuning helped the model learn process-specific input-output patterns and function like a regression model. Prompts were designed both for single-output and multi-output cases to assess predictive capacity (Table S3). The dataset was normalized and split into 80 % for training and 20 % for testing. The training loss curve (Fig. S3) showed stable, low loss, confirming proper convergence. The fine-tuned

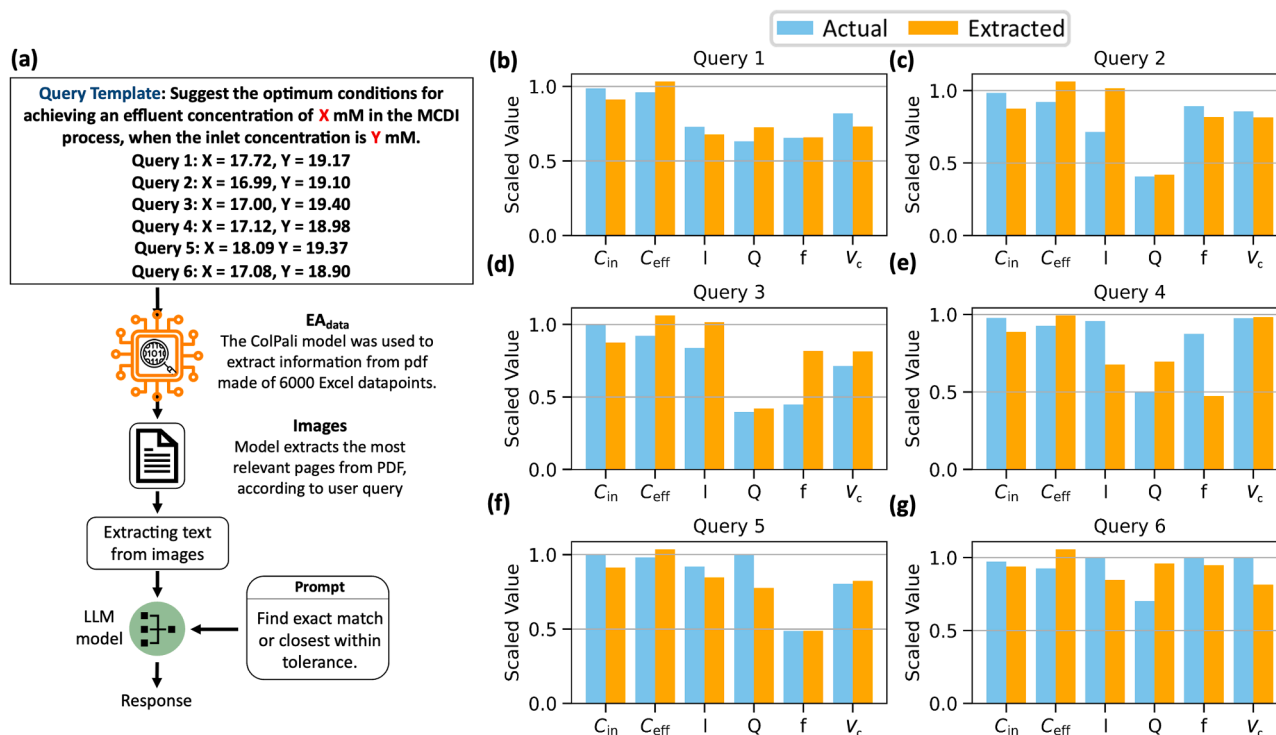


Fig. 3. Query-based evaluation of the expert-level agent for structured data (EA_{data}) in retrieving optimum operating conditions for MCDI processes. (a) Workflow and query template. Six representative user queries specify inlet and target effluent concentrations. The ColPali model extracts information from a PDF containing 6000 Excel data points, retrieves the most relevant pages, extracts text from images, and uses an LLM model with a prompt to find exact or closest matches within tolerance. (b–g) Comparison of **actual** and **extracted** values for key operating variables across Queries 1–6. Variables include inlet concentration (C_{in}), effluent concentration (C_{eff}), current (I), flow rate (Q), flushing intensity (f), and cell voltage (V_c). Scaled values near unity indicate accurate extraction. EA_{data} consistently reproduces concentration parameters with high fidelity, with minor deviations in secondary operating variables.

Mistral model performance was evaluated using regression plots and statistical metrics. Prompt 1 predicted effluent concentration, while Prompt 2 targeted current, flow rate, voltage, and flushing intensity. Prompts 3 and 4 addressed specific energy consumption (SEC) and thermodynamic efficiency (TEE). Predictions for effluent concentration, SEC, and TEE aligned closely with the one-to-one line, and both training and testing R^2 values exceeded 0.73 (Fig. 4a1, a6–a8, Table S4). Prompt 2 showed limited predictive accuracy; regression plots were scattered with negative R^2 values, reflecting difficulty in capturing weakly correlated variables (Fig. 4a2–a5, Table S4).

To address this limitation, a hybrid approach was developed in which Mistral model predictions for Prompt 2 were refined by a gradient boosting regressor. This integration improved performance substantially: testing R^2 for current rose from 0.03 to 0.63, and for flow rate from 0.01 to 0.61, while R^2 for voltage and flushing intensity also became positive (Fig. 4b1–b5, Table S5). These findings demonstrate that coupling a large language model with a standard machine learning model enhances its ability to connect broad pattern recognition with precise numerical prediction.

It is important to note that EA_{hybrid} is designed as a multi-prompt scientific framework rather than a single numerical prediction model. The fine-tuned language model can handle different types of MCDI prompts, including effluent concentration prediction, optimal operating condition suggestion, and energy performance estimation, where each prompt requires a different set of input variables and produces different outputs. If a traditional machine learning model such as gradient boosting regression (GBR) were used alone, separate models would be required for each prompt, and each model would need independent training and optimization, making the system more complex and less flexible.

A standalone GBR was therefore trained using the same dataset and the same input variables as Prompt 2 (effluent concentration and inlet

concentration) to provide a direct comparison (Table S5). The results show that GBR can achieve similar numerical prediction performance for Prompt 2 because the hybrid framework already uses GBR as a numerical refinement module. However, the standalone GBR is limited to structured numerical prediction and cannot handle multiple prompts or flexible input structures. In contrast, EA_{hybrid} combines the flexibility of the language model with the numerical accuracy of GBR, enabling the system to handle multiple scientific tasks within a unified and scalable framework.

Notably, EA_{hybrid} achieves $R^2 > 0.80$ for C_{eff} , SEC, and TEE against numerical model benchmarks (Fig. 5a), while also maintaining predictions under missing-input conditions (Fig. 5c1), a capability fundamentally unavailable to the physics-based model (or first-principles model) and standard ML model, which requires a complete set of inputs to solve the governing equations. The hybrid model was then tested as a controller for maintaining water quality. For effluent concentration targets of 17.0 mM and 16.5 mM, the model consistently tracked set-points with high stability, deviating slightly at transition points when the inlet concentration changed (Fig. 5b2). Corresponding manipulated variable variations are shown in Fig. S4. This behavior confirms strong predictive accuracy and adaptability to rapid operating shifts, supporting real-time monitoring and control applications where stable effluent quality is essential. Furthermore, the effect of data precision was also examined (Fig. S5). Training with four-decimal-point data produced smoother trends and minor accuracy gains, as the model better captured subtle variations. However, the improvement was small, and therefore two-decimal-point data were preferred for efficiency and interpretability. This comparison confirmed that model performance remains stable under varying data precision.

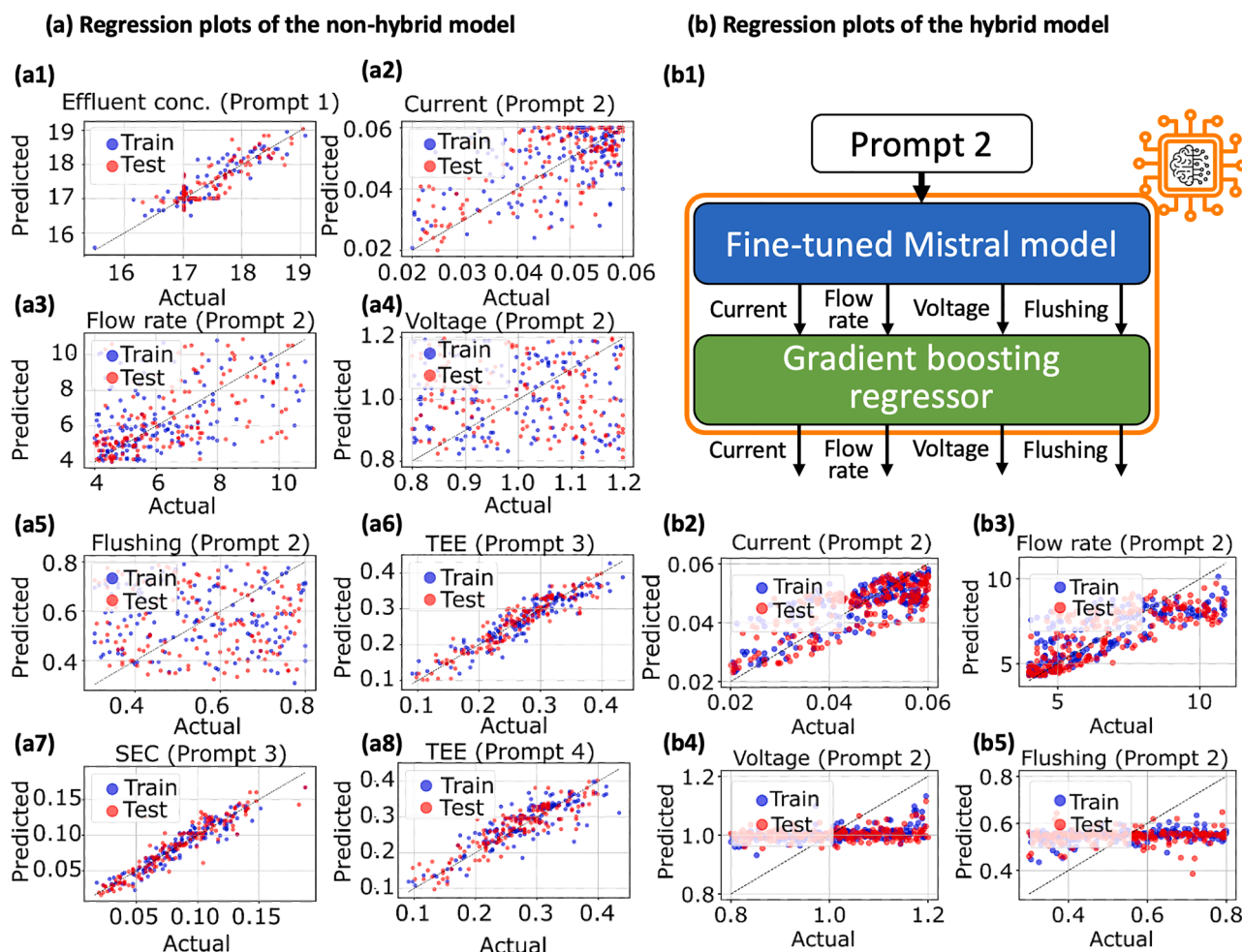


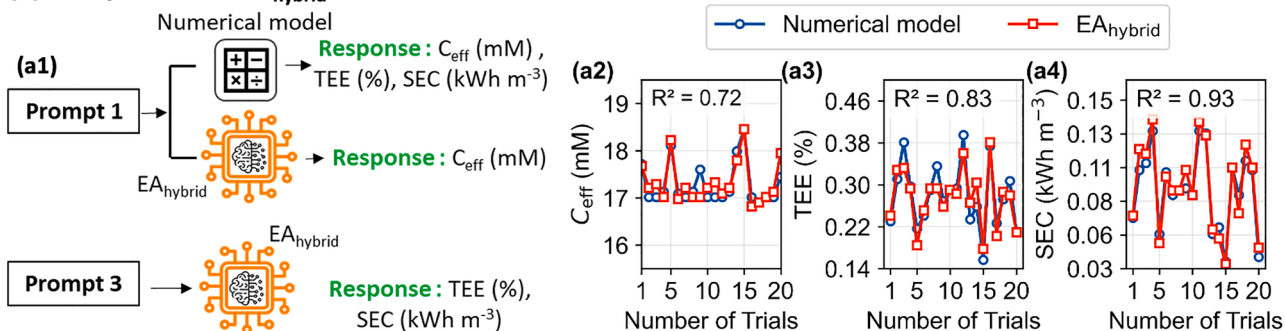
Fig. 4. Comparison of regression performances between non-hybrid and hybrid frameworks of the fine-tuned Mistral model (EA_{hybrid}). (a) Non-hybrid fine-tuned Mistral model performance. (a1) Effluent concentration prediction (Prompt 1) shows strong alignment with the one-to-one line ($R^2 > 0.73$ for both training and testing). (a2–a5) Operating condition predictions (Prompt 2) for current, flow rate, voltage, and flushing intensity show poor correlation with scattered data and negative R^2 values. (a6–a8) Thermodynamic energy efficiency (TEE, Prompts 3 and 4) and specific energy consumption (SEC, Prompt 3) demonstrate good predictive accuracy. (b) Hybrid model architecture and performance. (b1) Schematic showing the hybrid approach where fine-tuned Mistral model predictions for Prompt 2 are refined by a gradient boosting regressor for each operating variable. (b2–b5) Regression plots of the hybrid model for Prompt 2 variables show substantial improvement: current R^2 increases from 0.03 to 0.63, flow rate from 0.01 to 0.61, with voltage and flushing intensity also achieving positive R^2 values. Prompt descriptions: Prompt 1 predicts effluent concentration from operating variables; Prompt 2 suggests optimal operating conditions for target effluent quality; Prompt 3 predicts TEE and SEC; Prompt 4 estimates energy efficiency from effluent concentration and energy use parameters.

3.4. The hybrid model (EA_{hybrid}) can handle missing data input previously impossible for conventional models

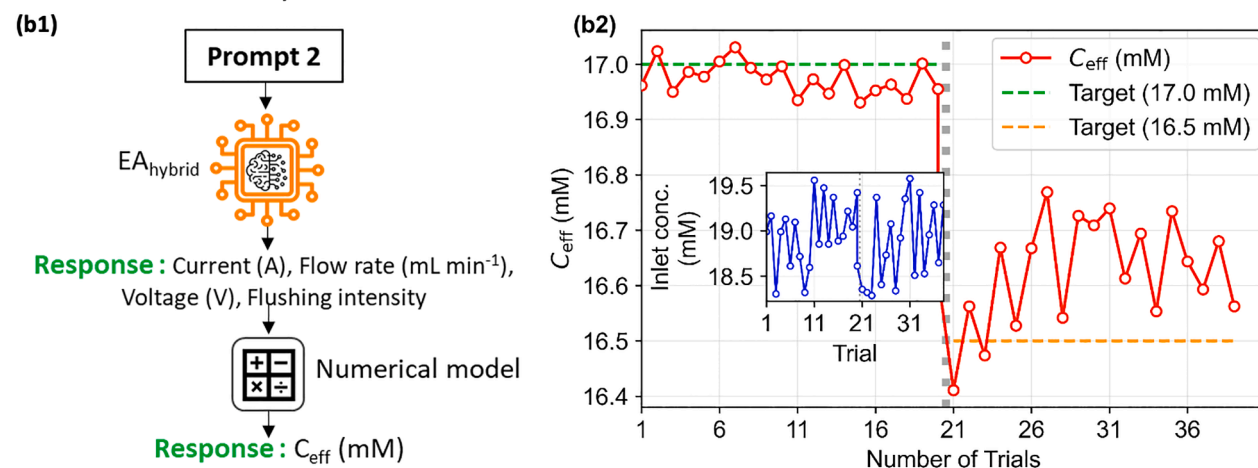
The resilience of EA_{hybrid} was tested under missing-input conditions. Remarkably, effluent concentration could still be predicted with reasonable accuracy even when one of the key inputs—current, voltage, flow rate, or flushing intensity—was omitted. Such performance is typically impossible for conventional machine learning models. The absence of flow rate or flushing intensity produced the largest deviations, whereas missing current or voltage caused smaller shifts (Fig. 5c1). Additional analysis with four-decimal-point data (Fig. S6) yielded slightly smoother trends and marginal accuracy gains; however, two-decimal-point data were retained for simplicity and interpretability, suggesting that the model is minimally sensitive to data precision. These results highlight both the relative importance of process inputs and the robustness of EA_{hybrid} under incomplete information, a frequent challenge in real-world water treatment systems where sensor noise or failures occur. To further evaluate the behavior of EA_{hybrid} under more challenging incomplete-input conditions, additional tests were conducted by removing multiple input variables simultaneously. Effluent

concentration predictions were evaluated for 10 representative trials under single, double, and triple missing input scenarios, and the results are summarized in Table S6. The average deviation from actual target values increased gradually from 0.22 mM (full input) to 0.39 mM (one missing input), 0.42 mM (two missing inputs), and 0.51 mM (three missing inputs), indicating a controlled and stable degradation in prediction accuracy as available information decreased. This behavior demonstrates that EA_{hybrid} can still provide reliable estimates even under severe incomplete-input conditions, highlighting the reasoning capability of the language model and the robustness of the hybrid framework in handling real-world situations where multiple sensors or process measurements may be unavailable. Together, the fine-tuned language model performs strongly for effluent concentration and energy metrics, while the hybrid approach extends its accuracy to less reliably predicted process variables. The EA_{hybrid} framework achieves accuracy comparable to numerical simulations and maintains resilience under imperfect conditions, offering a reliable and efficient surrogate for real-time monitoring and optimization of capacitive deionization systems.

(a) Comparison of EA_{hybrid} and numerical model



(b) Assessment of EA_{hybrid} performance as a controller



(c) Assessment of EA_{hybrid} performance under incomplete information

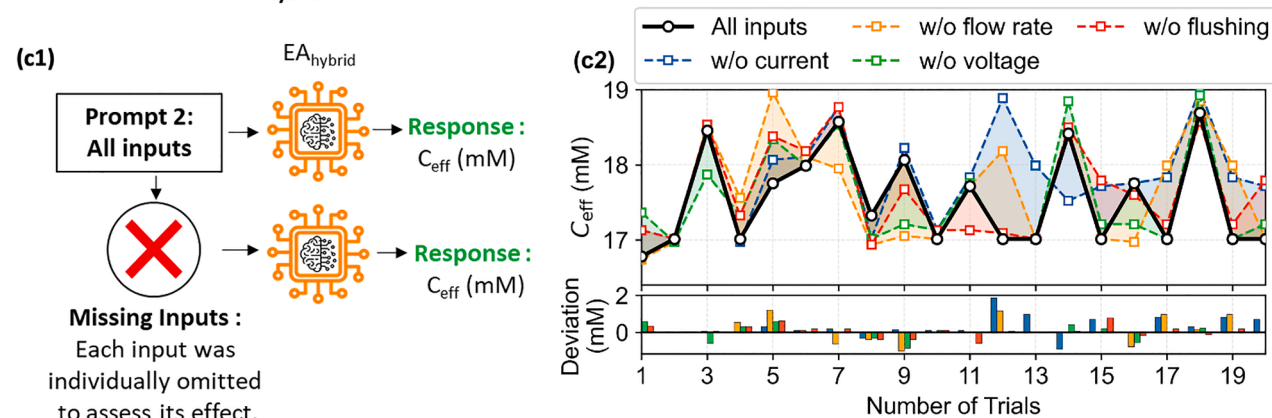


Fig. 5. Performance evaluation of the hybrid framework combining fine-tuned Mistral model with gradient boosting regressor (EA_{hybrid}). (a) Comparison with numerical model. (a1) Workflow showing EA_{hybrid} using an experimentally validated numerical model to generate responses for effluent concentration (C_{eff} , Prompt 1) and energy metrics (thermodynamic energy efficiency, TEE; and specific energy consumption, SEC, Prompt 3). (a2–a4) Time-series comparison across 20 trials shows strong agreement between EA_{hybrid} predictions (red) and numerical model results (blue) for C_{eff} , SEC, and TEE. (b) Controller performance assessment. (b1) Workflow for control application of EA_{hybrid} using Prompt 2 to predict operating conditions (current, flow rate, voltage, flushing intensity), which are fed to the numerical model to obtain C_{eff} . (b2) EA_{hybrid} maintains effluent concentration near target setpoints of 17.0 mM and 16.5 mM across 36 trials, with stable tracking despite inlet concentration fluctuations (inset shows inlet concentration variation for trials 1–31). (c) Robustness under missing inputs. (c1) Test configuration where individual inputs (current, voltage, flow rate, or flushing intensity) are omitted one at a time. (c2) Effluent concentration predictions under complete and incomplete information across 20 test cases. Missing information causes noticeable deviations (top panel), but the magnitude for each missing-input scenario is contained within 1–2 mM of deviation (bottom panel), confirming the model's resilience to incomplete data.

3.5. Cross-platform deployment of on-device AI agents using executable files

To demonstrate platform-independent standalone operation, the Python-based MCDI AI agent was converted into executable files for

Windows, Linux, and Raspberry Pi using auto-py-to-exe (Fig. 6a). Detailed installation and usage steps are provided in Text S3. On Windows and Linux, all three expert agents—EA_{know} (trained on 320 papers for theoretical analysis), EA_{data} (~6000 structured numerical data points), and EA_{hybrid} (for prediction and automation)—are fully

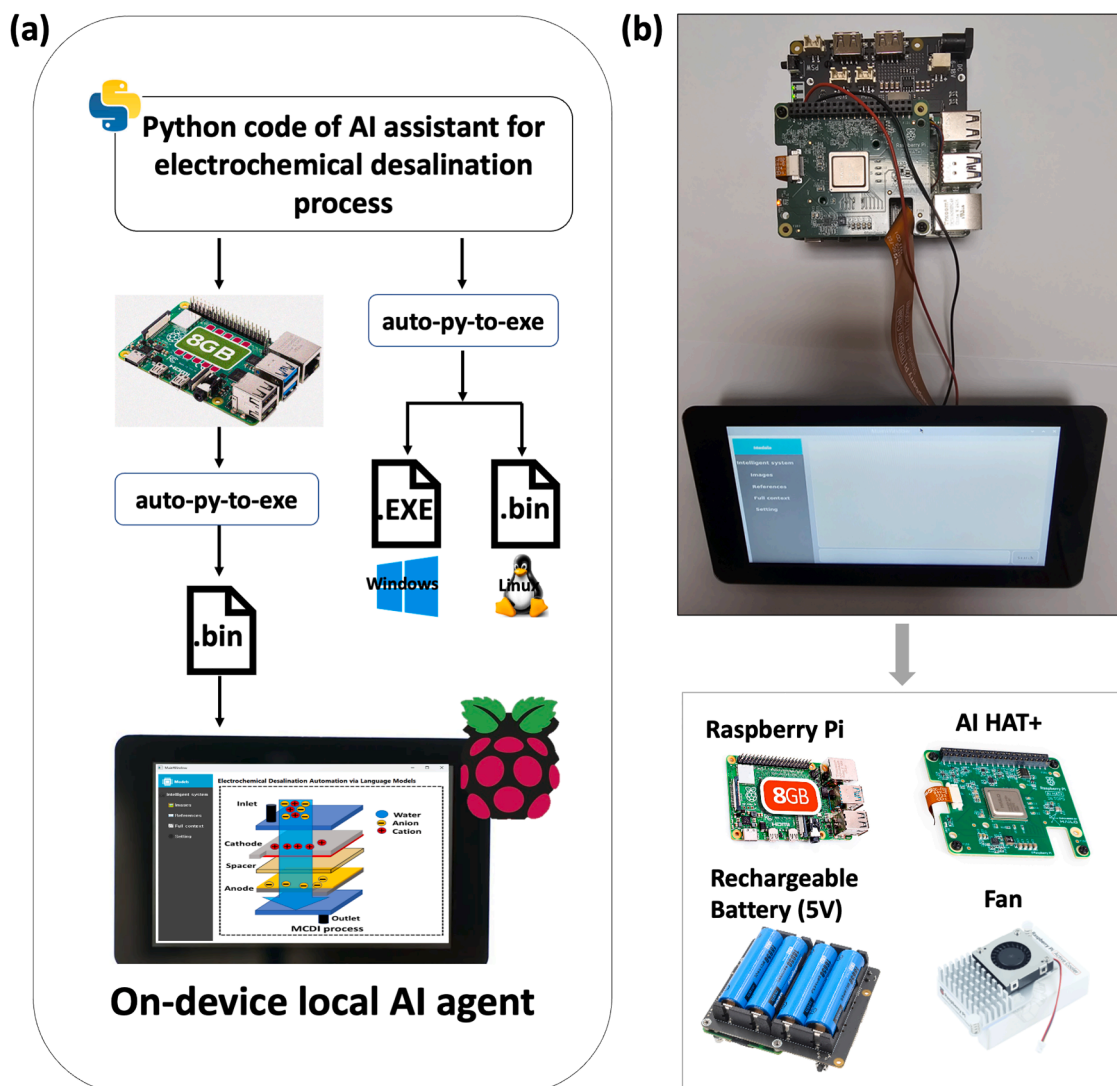


Fig. 6. Deployment and user interface of the on-device AI agent for electrochemical desalination. (a) Workflow for converting the Python-based AI agent into cross-platform executable files. The code is compiled using auto-py-to-exe to generate platform-specific binaries (.exe for Windows, .bin for Linux/Raspberry Pi), enabling standalone operation without Python installation. The resulting on-device AI agent provides a graphical user interface for accessing all three expert system components (EA_{know}, EA_{data}, and EA_{hybrid}). (b) Hardware deployment on Raspberry Pi 5. Top: Assembled system showing the Raspberry Pi 5 with AI HAT+ module and 7-inch touchscreen displaying the software interface. Bottom: Individual hardware components — Raspberry Pi 5, AI HAT+, rechargeable battery pack (5 V) for portable operation, and cooling fan. The system operates fully offline, providing on-site process optimization without cloud connectivity.

available. For Raspberry Pi, compact “Lite” versions were deployed due to limited hardware capacity: EA_{know} Lite (100 papers) used compact LLMs such as TinyLlama (638 MB), Gemma3:1B (815 MB), and SmolLM (990 MB); EA_{data} Lite used 1000 data points, and EA_{hybrid} Lite was quantized from 13 GB to 2 GB. The setup included an 8 GB RAM Raspberry Pi with a 5 V UPS (SupTronics V1.1), AI HAT+, and 7-inch touchscreen (Fig. 6b). EA_{know} Lite achieved performance comparable to foundation models such as ChatGPT, Claude and Le Chat (Fig. 7a). It showed slightly lower basic accuracy (0.28 vs 0.34–0.39) but comparable theoretical reasoning (0.43) and similar factual consistency (0.24–0.30) and conciseness (2.4–3.0) (Fig. 7a1–a4). The hallucination rate remained comparable to other models (≈ 0.84). EA_{data} Lite accurately predicted optimum operating conditions for specific inlet–effluent concentration pairs (Fig. 7b). Across four representative queries, scaled variables (C_{in} , C_{eff} , I , Q , f , V_c) remained near unity, confirming strong alignment between predicted and actual values and preservation of the full model’s optimization ability. The EA_{hybrid} Lite system was tested for real-time process control and adaptability (Fig. 7c). The system maintained effluent concentration near 17.0 mM and 16.5 mM setpoints

despite inlet fluctuations (Fig. 7c1), remained functional under missing-input conditions (Fig. 7c2), and produced thermodynamic efficiency (TEE) trends comparable to numerical simulations (Fig. 7c3). These results indicate that the quantized “Lite” systems retain the interpretive, predictive, and control capabilities of the full models while enabling fully on-device, portable AI deployment—a practical solution for laboratory and field environments lacking internet or cloud access.

4. Conclusion

This study develops and validates a portable, on-device artificial intelligence platform that unifies scientific literature reasoning, structured data analysis, prediction, and real-time process control for electrochemical water treatment. Using membrane capacitive deionization as a representative case, three expert agents — EA_{know}, EA_{data}, and EA_{hybrid} — were designed and systematically evaluated.

EA_{know} reduced hallucination rates by approximately 60 % compared to foundation models while maintaining 100 % reference attribution accuracy. EA_{data} retrieved key operating parameters from

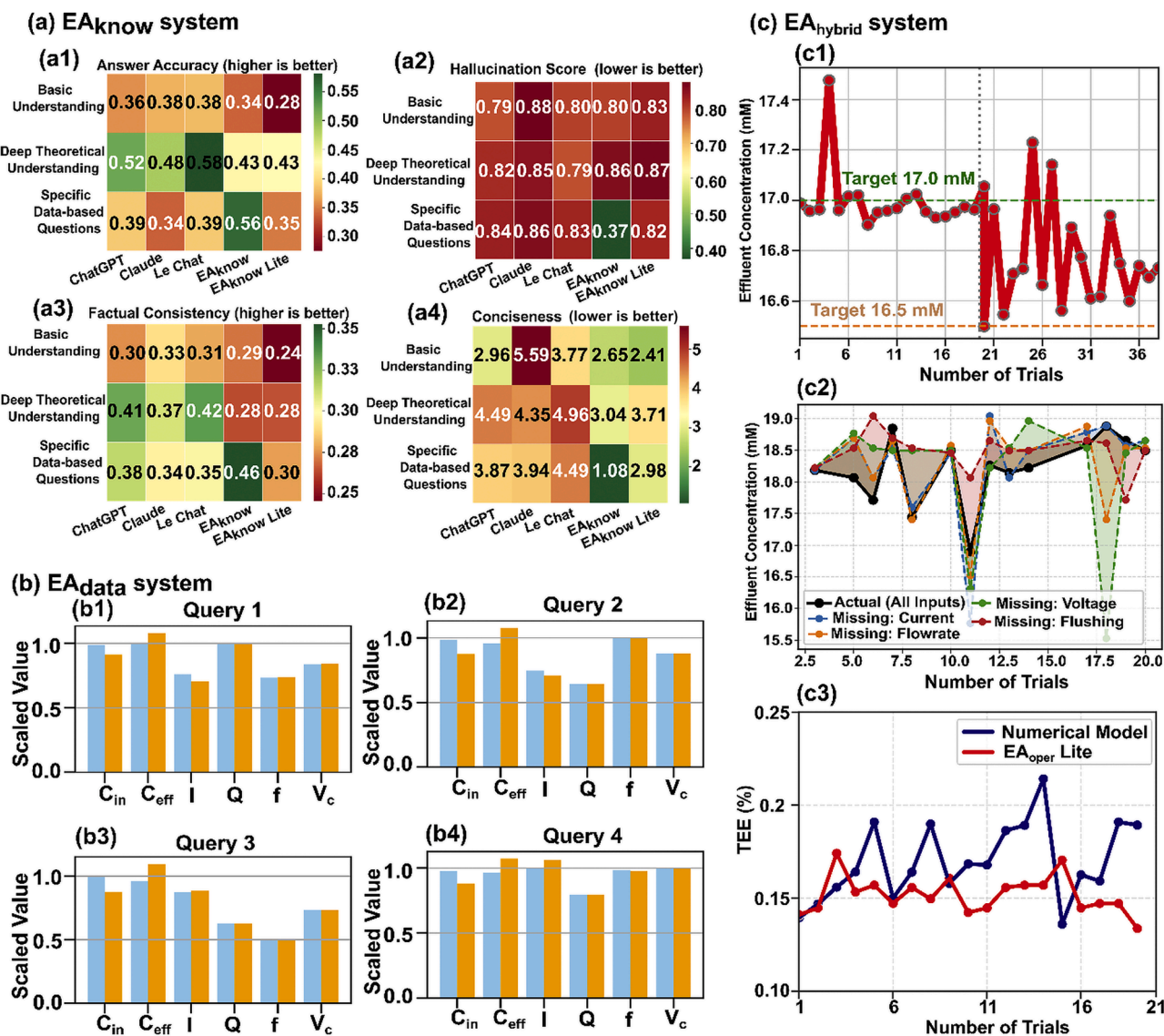


Fig. 7. Comparative performance of lightweight expert-level assistants (EA_{know} Lite, EA_{data} Lite, and EA_{hybrid} Lite) designed for resource-limited deployment on Raspberry Pi. (a) EA_{know} Lite system performance. (a1–a4) Heatmaps comparing answer accuracy, hallucination score, factual consistency, and conciseness against GPT-5.1 (ChatGPT), Claude Sonnet 4, Le Chat (November 2024 version), and the full EA_{know} system across three question categories (basic understanding, deep theoretical understanding, and specific data-based questions). EA_{know} Lite, trained on 100 papers using compact LLMs (TinyLlama, Gemma3:1B, SmolLM), achieves comparable performance to foundation models despite reduced corpus size. (b) EA_{data} Lite system performance. (b1–b4) Scaled variable comparisons for four representative queries extracting optimum operating conditions. Actual (blue) and extracted (orange) values for inlet concentration (C_{in}), effluent concentration (C_{eff}), current (I), flow rate (Q), flushing intensity (f), and cell voltage (V_c) remain near unity, confirming accurate retrieval from the reduced dataset of 1000 data points. (c) EA_{hybrid} Lite system performance. (c1) Control tracking of effluent concentration at 17.0 and 16.5 mM setpoints across 36 trials, maintaining stability despite inlet fluctuations. (c2) Robustness under missing inputs across 20 test cases, showing resilience when individual parameters are omitted. (c3) Thermodynamic energy efficiency (TEE) comparison between numerical model (blue) and EA_{hybrid} Lite (red) across 20 trials, demonstrating consistent predictive capability. The quantized EA_{hybrid} Lite model (reduced from 13 GB to 2 GB) retains the core functionality of the full system while enabling deployment on edge devices.

structured datasets with deviations below 5 %. Most importantly, EA_{hybrid} integrated language-based reasoning with machine learning to enable robust prediction and closed-loop control. It maintained effluent concentration within ± 0.2 mM of the target under dynamic inlet fluctuations and remained operational even when key sensor inputs were missing, increasing the R^2 for applied current prediction from 0.03 to 0.63. While the present framework is validated against a physics-based model incorporating fouling dynamics, direct validation on a physical MCDI cell, particularly under varying water chemistries and sustained operation, represents the essential next step. Such experimental validation is currently underway and will be the subject of a subsequent study. The multi-ion transport framework employed here (Na^+ , Ca^{2+} , Cl^-) provides a foundation for extending to more complex water matrices;

incorporating organic fouling descriptors and additional ionic species (e.g., Mg^{2+} , SO_4^{2-}) represents a direct pathway to improve applicability to diverse real-world water sources.

Deployment on a Raspberry Pi 5 confirmed fully offline operation in portable configurations, demonstrating feasibility for decentralized and resource-limited facilities. The modular framework can be extended to other electrochemical water treatment technologies, providing a scalable and explainable pathway toward autonomous water quality management.

Data availability

All data supporting the findings of this study, along with the original

code, are publicly accessible in the Zenodo repository under the DOI: <https://zenodo.org/uploads/17562436>.

Code availability

The codes used in this study are openly available in the Zenodo repository under the DOI: <https://zenodo.org/records/19343633>.

CRediT authorship contribution statement

Zahid Ullah: Writing – original draft, Visualization, Software, Methodology, Investigation. **Hoo Hugo Kim:** Writing – review & editing, Visualization. **Moon Son:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.watres.2026.125995](https://doi.org/10.1016/j.watres.2026.125995).

References

- Agarwal, S., Sahu, G., Puri, A., Laradji, I., Dvijotham, K., Stanley, J., Charlin, L., Pal, C., 2024a. LitLLM: A Toolkit For Scientific Literature Review. Cornell University. <https://doi.org/10.48550/arxiv.2402.01788>.
- Agarwal, S., Sahu, G., Puri, A., Laradji, I., Dvijotham, K., Stanley, J., Charlin, L., Pal, C., 2024b. LitLLMs, LLMs For Literature Review: Are we There yet? Cornell University. <https://doi.org/10.48550/arxiv.2412.15249>.
- Alkhadra, M., Su, X., Suss, M.E., Tian, H., Guyes, E.N., Shocron, A.N., Conforti, K.M., De Souza, J.P., Kim, N., Tedesco, M., Khoiruddin, K., Wenten, I.G., Santiago, J.G., Hatton, T.A., Bazant, M.Z., 2022. Electrochemical methods for water purification, ion separations, and energy conversion. *Chem. Rev.* 122, 13547–13635. <https://doi.org/10.1021/acs.chemrev.1c00396>.
- Barazesh, J., Hennebel, T., Jasper, J.T., Sedlak, D.L., 2015. Modular advanced oxidation process enabled by cathodic hydrogen peroxide production. *Env. Sci. Technol.* 49, 7391–7399. <https://doi.org/10.1021/acs.est.5b01254>.
- Bednarczyk, L., Reichenpfer, D., Gaudet-Blavignac, C., Ette, A.K., Zagher, J., Zheng, Y., Bensahla, A., Bjelogrić, M., Lovis, C., 2025. Scientific evidence for clinical text summarization using large language models: scoping review. *J. Med. Internet Res.* 27, e68998. <https://doi.org/10.2196/68998>.
- Chen, M., Shao, W., Xu, P., Wang, J., Gao, P., Zhang, K., Luo, P., 2025. EfficientQAT: efficient quantization-aware training for large language models. *Assoc. Comput. Linguist.* 10081–10100. <https://doi.org/10.18653/v1/2025.acl-long.498>.
- Faysse, M., Sibille, H., Wu, T., Omrani, B., Viaud, G., Hudelot, C., Colombo, P., 2024. ColPali: efficient document retrieval with vision language models. *arXiv*. <https://doi.org/10.48550/arxiv.2407.01449>.
- Guyes, E.N., Shocron, A.N., Chen, Y., Diesendruck, C.E., Suss, M.E., 2021. Long-lasting, monovalent-selective capacitive deionization electrodes. *NPJ Clean Water* 4, 22. <https://doi.org/10.1038/s41545-021-00109-2>.
- He, Z., Miller, C.J., Zhu, Y., Wang, Y., Fletcher, J., Waite, T.D., 2024. Membrane capacitive deionization (MCDI): a flexible and tunable technology for customized water softening. *Water Res.* 259, 121871. <https://doi.org/10.1016/j.watres.2024.121871>.
- Jiang, A., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D., Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L., Lachaux, M.A., Stock, P., Scao, T., Lavril, T., Wang, T., Lacroix, T., Sayed, W., 2023. Mistral 7B. *arXiv*. <https://doi.org/10.48550/arxiv.2310.06825>.
- Kim, H.H., Choi, B., Ullah, Z., Jeong, N., Cho, K.H., Park, S., Baek, S.S., Son, M., 2024. Decoupling ion concentrations from effluent conductivity profiles in capacitive and battery electrode deionizations using an artificial intelligence model. *Water Res.* 262, 122092. <https://doi.org/10.1016/j.watres.2024.122092>.
- Kim, N., Lee, E.A., Su, X., Kim, C., 2021. Parametric investigation of the desalination performance in multichannel membrane capacitive deionization (MC-MCDI). *Desalination* 503, 114950. <https://doi.org/10.1016/j.desal.2021.114950>.
- Kommineni, V.K., Ahmed, W., Koenig-Ries, B., Samuel, S., 2024a. Automating information retrieval from biodiversity literature using large language models: a case study. *Biodivers. Inf. Sci. Stand.* 8. <https://doi.org/10.3897/biss.8.136735>.
- Kommineni, V., König-Ries, B., Samuel, S., 2024b. Harnessing Multiple LLMs For Information Retrieval: A case Study On Deep Learning Methodologies in Biodiversity publications. *Arxiv*. Cornell University. <https://doi.org/10.48550/arxiv.2411.09269>.
- Lee, S., Shim, J., Kim, H.H., Yun, N., Son, M., Cho, K.H., 2025. Optimizing capacitive deionization operation using dynamic modeling and reinforcement learning. *Desalination* 602, 118626. <https://doi.org/10.1016/j.desal.2025.118626>.
- Łukaszczuk, Z., Burda, K., 2024. Energy efficiency of a water and wastewater company – CASE study. *Sci. Pap. Sil. Univ. Technol. Organ. Manag. Ser.* 2024, 253–265. <https://doi.org/10.29119/1641-3466.2024.212.15>.
- Neijjar, M., Zacharias, L., Stiehle, F., Weber, I., 2023. LLMs for science: usage for code generation and data analysis. *arXiv*. <https://doi.org/10.48550/arxiv.2311.16733>.
- Poornima, A., Agarwal, R., Simha, J.B., 2024. Automated Literature Review Using Large Language Models. Springer Nature, Singapore, pp. 415–425. https://doi.org/10.1007/978-981-97-3526-6_32.
- Sarker, I.H., 2021. Machine Learning: algorithms, real-world applications and research directions. *SN. Comput. Sci.* 2, 160. <https://doi.org/10.1007/s42979-021-00592-x>.
- Scherbakov, D., Hubig, N., Jansari, V., Bakumenko, A., Lenert, L.A., 2025. The emergence of large language models as tools in literature reviews: a large language model-assisted systematic review. *J. Am. Med. Inform. Assoc.* 32, 1071–1086. <https://doi.org/10.1093/jamia/ocaf063>.
- Shi, T., Ding, Y., 2025. Systematic characterization of LLM quantization: a performance, energy, and quality perspective. *arXiv*. <https://doi.org/10.48550/arxiv.2508.16712>.
- Thirunavukarasu, A.J., Ting, D.S.J., Elangovan, K., Gutierrez, L., Tan, T.F., Ting, D.S.W., 2023. Large language models in medicine. *Nat. Med.* 29, 1930–1940. <https://doi.org/10.1038/s41591-023-02448-8>.
- Ullah, Z., Yun, N., Rossi, R., Son, M., 2025. Autonomous water quality management in an electrochemical desalination process. *Water Res.* 280, 123521. <https://doi.org/10.1016/j.watres.2025.123521>.
- Xu, B., Li, Z., Yang, Y., Wu, G., Wang, C., Tang, X., Li, Y., Wu, Z., Su, Q., Shi, X., Yang, Y., Tong, R., Wen, L., Ng, H.Y., 2025a. Evaluating and advancing large language models for water knowledge tasks in engineering and research. *Env. Sci. Technol. Lett.* 12, 289–296. <https://doi.org/10.1021/acs.estlett.5c00038>.
- Xu, B., Wu, G., Li, Z., Xu, G., Zeng, H., Tong, R., Ng, H.Y., 2025b. Towards domain-adapted large language models for water and wastewater management: methods, datasets and benchmarking. *NPJ. Clean. Water.* 8, 82. <https://doi.org/10.1038/s41545-025-00509-8>.
- Xu, L., Liu, W., Zhang, X., Tang, W., Lee, D.J., Yang, Z., Chen, Z., Lin, S., 2024. Pseudo-continuous and scalable electrochemical ion pumping with circuit-switching-induced ion shuttling. *Nat. Water* 2, 999–1008. <https://doi.org/10.1038/s44221-024-00312-8>.
- Yoon, N., Park, S., Shim, J., Lee, J., Son, M., Cho, K.H., 2022a. Membrane capacitive deionization model including fouling indexes obtained via real-time fouling layer measurements. *Desalination* 536, 115852. <https://doi.org/10.1016/j.desal.2022.115852>.
- Yoon, N., Park, S., Son, M., Cho, K.H., 2022b. Automation of membrane capacitive deionization process using reinforcement learning. *Water Res.* 227, 119337. <https://doi.org/10.1016/j.watres.2022.119337>.
- Zhu, Y., Miller, C., Lian, B., Wang, Y., Fletcher, J., Zhou, H., He, Z., Lyu, S., Purser, M., Juracich, P., Sweeney, D., Waite, T.D., 2024. Brackish groundwater desalination by constant current membrane capacitive deionization (MCDI): results of a long-term field trial in Central Australia. *Water Res.* 254, 121413. <https://doi.org/10.1016/j.watres.2024.121413>.